Natalia A. Agapova

ML Engineer

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Education

Innopolis, Russia **Innopolis University** 2022 – 2026

- **Major:** BSc in Applied Artificial Intelligence (GPA: 4.02 / 5)

Employment

Croissan Studio **Apr 2025 to present**

ML Software Engineer *Apr 2026 to present*

- Work as the ML engineer in client teams delivering end-to-end AI products, alongside designers and software engineers.
- Fully fine-tuned transformer language models and adapted others through prompting for client products.
- Compared model versions on task metrics to choose what goes into the product.
- Trained and adapted generative image and talking-avatar models for client projects.

ML Engineer & Product Designer *Jul 2025 to Apr 2026*

- Reworked the core model behind NeuroArtist, with the freedom to explore and test my own approach.
- Iterated on model outputs against client-facing quality bars, running comparisons to decide what shipped.
- Also led the NeuroArtist rebrand across physical and digital materials, and contributed to the Croissan Studio website redesign.

Software Engineer Intern *Apr 2025 to Jul 2025*

- Worked on the NeuroArtist website.
- Helped with ML tasks on the team, and was offered an ML Engineer role after the internship.
- Contributed to client projects, for example a tourism website and the Moscow 2030 event, etc.

- **Leveraged knowledge** in Python, PyTorch, Hugging Face Transformers, Diffusers, and LoRA fine-tuning.
- *Letter of recommendation from the studio's CEO* →

Asimov Lab (startup) **Feb to Sep 2024**

Frontend Team Lead & ML Developer

- Led the frontend team to the MVP of an AI platform that generates tests for teachers and students.
- Built the MVP version of the website end to end.
- One of six ML developers behind the core model of the product.

- **Leveraged knowledge** in React, TypeScript, Python, and scikit-learn.

Freelance **2022 to present**

ML Engineer

- Applied ML work for independent clients.

- **Leveraged knowledge** in Python, PyTorch, and scikit-learn.

Personal Projects

Fair Resume Screening NLP · Explainability · Debiasing · Published, first author

- Built a BERT model that sorts resumes into 9 IT job categories using real HeadHunter data, then audited it for bias and tested 39+ ways to fix it.
- Found that swapping just the applicant's city flipped the model's decision in 7.7% of cases (baseline accuracy 60.9%) – published this bias finding as first author.
- **Utilized:** Python, PyTorch, BERT, Captum, pandas.

Emotion Detection Computer Vision · On-device Deployment

- Trained a CNN to recognize emotions from faces, reaching 58.7% accuracy (up to 60.8% with ensembling), then shrank it to run on-device with just 1.15 ms latency (Core ML, ONNX Runtime Web).
- **Utilized:** Python, PyTorch, Core ML, ONNX Runtime.

Personal Knowledge System RAG · NLP · Embeddings

- Built a personal AI assistant that answers only from your own PDFs and always cites the exact file and page, powered by a local LLM (Llama 3.2) so nothing leaves your machine.
- **Utilized:** Python, SentenceTransformer, ChromaDB, Ollama.

Skills

ML / AI: *(proficient)*: Python, PyTorch, BERT, Captum, model evaluation & debiasing, RAG (ChromaDB, SentenceTransformer) *(familiar)*: LLMs (Ollama), OpenCV, MediaPipe, Core ML, ONNX Runtime
Tools: *(familiar)*: Figma, front-end development, Git/GitHub
Languages: Russian C2 (native) · English C1+ · Ukrainian B1 · French A2 · Tatar A2 · Korean A1

Publications

Agapova, N. A.; Lukmanov, R. A. "Improving Fairness in AI-Powered Recruitment: An Interpretable Resume Screening System", 2026. doi:10.66693/mathai.1017

Kholodov, Y. A.; Salloum, H.; Agapova, N. A. "Quantum Annealing Approaches to Breaking RSA Encryption", *Voprosy kiberbezopasnosti*, 2025. doi:10.21681/2311-3456-2025-4-127-133